

Headphones

Get a head-full of personalised audio with one of these

In India, headphones have been something of a nascent product category. While it seems that cans have been around forever, the acceptance level has been very low. People either need cans, or don't. A home with a decent speaker system will rarely also need a headphone, unless someone wishes to listen to music without disturbing others, or a lot of chatting via the Internet takes place. If you give ten people a choice between speakers and headphones, nine would opt for the former. I've actually tried asking more than a hundred people, only two or three preferred headphones, and these were mainly gamers who lived with families and did not want to disturb them.

Most people need headphones for chatting with friends on the Internet. The next group are those who use portable multi-media players (PMPs), and 95 per cent of this bunch will make do with the headphones supplied with their PMPs. Some prefer to use headphones while gaming, if only to prevent others from being disturbed. A select few will actually prefer a headphone for the audio quality over a set of similarly priced speakers, and these are the self-proclaimed audiophiles.

So why opt for a headphone? Well, a good headphone will offer sound quality as good as any speaker system that costs nearly ten times as much. Just as any speaker unit needs an amplifier, certain audiophile grade headphones require headphone amplifiers to really drive their voice coils. I've come across audiophile grade headphones such as Sennheiser's HD650 and HD600, BeyerDynamics' DT880, AKG's K701 and Grado's RS 1 and RS2, and I am very impressed with their sonic abilities. Of course such headphones



and DAC combo, in case your source is a PC, to enjoy the most out of such headphones.

The fact is that headphones are as varied as any other product category, and it's safe to say that no two headphones are created alike. There are also a number of types of headphones, based on the technology they employ to produce sound, their size and fit on your head, and the sound quality they offer. As far as the technology goes, there are two types of headphones — Dynamic and Electrostatic. Dynamic are the most widely used and common; and they can be as cheap or as costly as you like.

The main technology here is of a stationary magnetic field and its interaction with the voice coil; this is attached to a moving, i.e. dynamic, diaphragm which is made of isobutyl rubber, polymer or cellulose-like material. The moving coil produces an alternating magnetic field which interacts with the static magnetic field, and this causes the diaphragm to move and therefore displace air, which creates sound. Electrostatic headphones consist of a thin charged diaphragm which is placed between two perforated metal electrodes. Depending on the polarity, the diaphragm moves towards one or the other plate creating sound.

We also classify headphones on the basis of their form factors. Circumaural headphones are the largest ones with large earcups that

fit around your ears, encompassing them, and they usually have a headband for holding them together for a snug fit around your ears. Typical audiophile headphones are usually circumaural, although this isn't necessary. They are considered by some to be the most comfortable, although this varies with the design. Supraaural headphones have smaller earcups — these sit on your ears and are considered to be the most uncomfortable. Earbuds are the ones which ship with most PMPs. These are small enough to fit inside your ear. Canal earphones or earplugs are equally small, and they actually protrude inside your ear canal, and usually have a soft inner material (generally silicone). These offer much better noise isolation compared to regular earbuds.

Basically, earbuds and canal type earphones are for you if you desire a portable setup, i.e., a PMP player, or something that you can easily carry with you. Earbuds offer less dynamic range of sound and noise isolation in comparison to earplugs which offer better sound quality, although they are costlier as well. Earplugs are very suitable for those wanting to listen to music or watch movies while aboard noisy public transport. Unfortunately, good earplugs are as rare as a dodo egg in our markets. Creative is the only brand offering affordable in-ear type earplugs, and their EP630 is immensely popular thanks to its coupling very good sound quality with an affordable price of Rs 800 or thereabouts. Alternative brands include audiophile heavyweights like Sennheiser and Bose. The latter has just one offering called the Bose Intra Ear, and these are a very neutral sounding set of earplugs that are suitable for even serious music lovers — they're that good. The only minus is their price — Rs 4,800. Sennheiser's PXC 500 is also available, and offer similar performance to Creative's EP630s for a much steeper price of Rs 2,500. If you are looking for a companion for your iPod, Cowon D2 or any such PMP, I suggest Creative's EP630 — terrific value for money. You might argue that your PMP or cellphone shipped with earphones. Well, dump these if you're even remotely looking

at a decent portable solution. The difference between these and even the entry level, (by audiophile standards), EP630 is very noticeable. I do not recommend earphones, given a choice between them and earplugs.

Active noise cancellation headsets are also available from Sennheiser and Bose. Basically, such headphones have an additional circuit present whose only function is to kill outside noise. Such headphones are useful when you are indoors and the ambient environment is noisy. Other headphones without this feature, but which isolate noise well are called passive noise cancellation phones. In my opinion however, active noise cancellation headsets aren't worth the price, and unless you're über rich I suggest you skip this category altogether. Bose's Quiet Comfort 3 is available for Rs 30,000, whereas the older Quiet Comfort 2 is available for Rs 25,000 — overpriced in my opinion, as they aren't as good as top-end cans should be. Sennheiser's PXC series is also available. The PXC450 is a good headphone although it needs a headphone amp to really shine. In fact, without an amp, the bass and mid-range is muddy and unrevealing, and the highs very harsh and unfocussed. For Rs 16,990, the PXC450 is a little better priced than the Bose offerings. This is way too much for such demanding cans (they need an amp remember?). For travel, or any sort of noisy environment, a set of EP630 earplugs will isolate noise nearly as well, and at a fraction of the price.

Supra-aural headphones are somewhat rarer. In fact, this design was made famous by Grado Labs, of Brooklyn, USA, and Grados aren't officially available in India, although there are some people who will be able to import them for you. Check out forums for such deals. There are some local manufacturers who model their headphones after such supra-aural designs, but beware — these are mostly horrible sounding. In fact, when dealing with sound, its best to go for trusted brands — I like to think of it as a one time investment, where you shell out big time once, but then have great sound at your beck and call for a long time to come.

If there's one category of headphones that available in droves, it's the circumaural variety. Most of the headphones available in our markets from the lowly Rs 250 ones to the ultra-high-end variety are circumaural. The reason is that such around-the-ear headphones in general do not cause as much discomfort as the other form factors. Brands like Philips, Creative and Sennheiser have some quality entry-level headsets which are priced around the Rs 1,200 mark and upwards. If you want *el cheapo* headphones then brands like iBall, Intex and Zebronic have a variety of models that should satisfy you. Sennheiser's HD210 and HD212 are decent entry level headphones priced at Rs 1,200 and Rs 1,400 respectively. Philips' SHP2000 is another stereo set available for Rs 1,600.

Now the big question — how much should you spend, and what exactly is suitable for you? The answer depends on what you expect from your headphones. If you're looking at something very basic for chatting — do not blow a lot of money. If you are looking for something decent for gaming, be prepared to shell out at least Rs 1,200 or thereabouts. There are also some surround-sound headsets available from the likes of Zalman. Their ZMRS6F are 5.1 surround-sound headphones, and are priced at Rs 2,800. These are ideal for gamers and home theatre buffs as they provide an earphone equivalent of a 5.1 surround system.

The fact is that most of the so called audiophile brands, i.e. Bose, Sennheiser and their ilk, are grossly overpriced in India. There are also brands such as AKG, Audio Technica and Grado which aren't officially available in India at all. There are some options available for those interested in high-end headphones, but don't wish to pay the premium.

As mentioned above, some people will import such items for you, for a small premium of course. These intrepid dealers also offer a warranty at their own risk, and for someone looking at saving anywhere between 40 to 200 per cent on the Indian price of the component, this is a good deal. The entry-level market is however chock-full of options, so, if you're on a shoestring budget, this is the right place to shop.